Collaborative Micro-productivity Project: Establishment-Level Productivity Dataset, 1972-2020

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Abstract

We describe the process for building the Collaborative Micro-productivity Project (CMP) microdata and calculating establishment-level productivity numbers. The documentation is for version 7 and the data cover the years 1972-2020. These data have been used in numerous research papers and are used to create the experimental public-use data product Dispersion Statistics on Productivity (DiSP).

[*] Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. This paper does not use any confidential Census Bureau data. The core Collaborative Micro-productivity Project also includes Cindy Cunningham, Lucia Foster, John Haltiwanger, Sabrina Pabilonia, and Jay Stewart.

1. Introduction

The Collaborative Micro-productivity Project (CMP) is a joint project with the Bureau of Labor Statistics that aims to enhance BLS' official aggregate productivity statistics with information from the Census Bureau's micro-level business data. In this paper, we describe the process for building the CMP microdata and calculating establishment-level productivity numbers.[1] The resulting dataset is used for (1) research into productivity dynamics by researchers at the Census Bureau and in the Federal Statistical Research Data Centers and (2) to create the experimental Dispersion Statistics on Productivity (available on the DiSP websites)[2]. A full list of papers related to CMP can be found on the DiSP Publications webpage.

The build sequence described in this paper creates two establishment-level productivity indicators (labor productivity and total factor productivity) using information from existing surveys collected by BLS and Census, as well as other data listed below. Labor productivity is defined as the ratio of deflated output and total hours worked. Total factor productivity is calculated as the ratio of deflated output and an input index where the index is a weighted geometric average of capital, total hours, intermediate inputs, and energy (and the weight of each input is given by the expenditure share of the input in each 6-digit NAICS industry's total cost). Details on CMP microdata and methodology can be found in Cunningham et al. (2022).

The data underlying the CMP microdata are listed below.

- Census Bureau Business Microdata
  - Annual Survey of Manufactures (ASM)
  - Census of Manufactures (CMF)[3]
  - Longitudinal Business Database (LBD)
- External Data[4]
  - NBER-CES Manufacturing Industry Database (NAICS and SIC)
  - BEA Industry Data (NAICS and SIC)
  - BLS Industry Data (NAICS and SIC)

To construct the CMP microdata, we first create a dataset of industry-level deflators and cost shares based on external sources. We then link the industry-level data to the ASM and CMF. We use the linked data to create establishment-level measures of inputs and output and calculate productivity measures.

The CMP microdata are available to qualified researchers on projects approved to use the ASM/CMF data through the Federal Statistical Research Data Centers (www.census.gov/fsrdc).

2. Combining Industry-Level External Data

The NBER-CES, BEA, and BLS industry-level external data are combined by industry and year. Combined SIC data are created for 1972-1997 and combined NAICS data are created for 1997 onwards. Note, we have one SIC-NAICS overlap year, 1997. Variables in both external datasets are converted to use a base-year of 1997 where appropriate (see the following section for details).

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[1] This note is publicly available and covers methodology only. A version of this note with more detail on the underlying data is available internally to qualified researchers on projects approved to access the underlying microdata (CMP Team, 2023).
[2] www.census.gov/disp and https://www.bls.gov/productivity/articles-and-research/dispersion-statistics-on-productivity/
[3] The CMF is part of the Economic Census and is a quinquennial survey that collects data for years ending in "2" and "7". In CMF years, the ASM is collected as part of the CMF. Observations based solely on administrative records data are excluded.
[4] The external data are described in detail in Appendix A.

2.1 SIC (1972-1997)

The 1972-1997 NBER-CES, BEA, and BLS industry-level external SIC data are combined by industry and year. The NBER-CES data are provided at the 4-digit SIC level. We use 1972 SIC-based NBER-CES data for 1972-1986 and 1987 SIC-based NBER-CES data for 1987-1997. The BEA data are provided with SIC-based industry descriptions. These are converted to SIC industry codes (roughly 2-digit SIC, but some 3-digit SIC groups). All BEA industry-level observations merge successfully to the NBER-CES database. BLS data are provided at the 2-digit SIC level. All BLS data merge successfully to the combined NBER-CES/BEA data.

2.2 NAICS (1997-2020)

The 1997-2018 NBER-CES[5] and 1997-2020 BEA and BLS industry-level external NAICS data are combined by industry and year. We use 1997 NAICS-based NBER-CES data for 1997-2011 and 2012 NAICS-based NBER-CES data for 2012-2018. The NBER-CES data are provided at the 6-digit NAICS level. The BEA data are provided with NAICS-based industry descriptions, which are converted to NAICS industry codes (roughly 3-digit NAICS, but some 4-digit NAICS groups). All BEA industry-level observations merge successfully to the NBER-CES database. The only BEA groups that are not merged to the data are the higher-level groupings (e.g., 31-33), which is expected; there is no reason to merge on the less specific data. BLS data are provided at the 3-digit NAICS level with some 3-digit industries combined. All BLS data merge successfully to the combined NBER-CES/BEA data.

3. Convert Industry-Level Data to a 1997 Basis

We convert our industry-level values to a 1997 basis because we have SIC-NAICS overlap in 1997.

3.1 Deflators

1972 SIC Years: 1972-1986

Deflators in the 1972 SIC NBER-CES database are given on a 1987 basis (1987=1). The complete 1972 SIC NBER-CES database only goes through 1996, the 1997-values of deflators are not available. The file contains 1997 values for *PIMAT* and *PISHIP*, but 1997 values for *PIEN* and *PIINV* are missing. *PIMAT* and *PISHIP* values are easily converted to a 1997 basis as shown in (1) for *PIMAT*.

$$PIMAT_{i,t} = \frac{PIMAT_1987B_{i,t}}{PIMAT_1987B_{i,1997}} \quad (1)$$

PIEN and *PIINV* for 1972 SIC industries on a 1997 basis are created using the 1987 SIC deflators (once converted to 1997=1 as discussed below) in conjunction with the 1972 SIC - 1987 SIC concordance file provided with the June 2000 version of the NBER-CES database.[6] The concordance contains four variables: *SIC87* (1987 4-digit SIC), *SIC72* (1972 4-digit SIC), *SH8772* (shipments share of the 1987

[5] Data from 2019 onward is not available, so we impute it as described in Section 3.

[6] The concordance file, *conch_sh.xls*, was downloaded from http://www.nber.org/nberces/nbprod96.htm (accessed November 5, 2012). The file is no longer available at that link, but a file called *conch_sic72_sic87* is downloadable from the most recent version of the NBER-CES database at https://www.nber.org/research/data/nber-ces-manufacturing-industry-database.

SIC industry in the 1972 SIC industry), and *SH7287* (shipments share of the 1972 SIC industry in the 1987 SIC industry). *PIEN* and *PIINV* on a 1997 basis are created as shown in Equation (2).

$$PIEN_{i,t} = \sum_{j \to i} SH7287_{j,t} * PIEN_{j,t} \quad (2)$$

1987 SIC Years: 1987-1997

Deflators in the 1987 SIC NBER-CES data –*PIEN*, *PIINV*, *PIMAT*, and *PISHIP* –are given on a 1987 basis. They are converted to 1997=1 as shown in Equation (1). However, there are some complications. The first one is for the Asbestos industry (SIC = 3292) where data is missing for 1992-1997. There are no ASM/CMF observations in the Asbestos industry after 1993. See Section 4 for details on how NBER-CES deflator values for the Asbestos industry are created for 1992-1997 to allow for conversion of Asbestos deflators to 1997=1. The second complication is that there are no 1997 values for SIC industries that move out of manufacturing with the switch to NAICS in 1997. See Section 4 for a list of these industries and for details on how 1997 NBER-CES deflator values are estimated for them to allow for conversions to 1997=1.

NAICS Years: 1997-2020

The deflators in the NBER-CES NAICS database – *PIEN*, *PIINV*, *PIMAT*, and *PISHIP* – are already given on a 1997 basis (1997=1) through 2011. From 2012-2018 (these years represent the most-updated version of the NBER-CES database), the deflators are on a 2012 basis (2012=1). To convert 2012 and later values to a 1997 basis, we use the 1997 values of the deflators taken from the 2012-basis dataset (for example, $PIEN_2012B_{i,1997}$). The deflators we obtain from BLS (*PIINVE* and *PIINVS*) are given on a 2012 basis (2012=1). They are converted to 1997=1 by dividing by the 1997 value of the deflator. The deflators we obtain from BEA are also given on a 2012 basis (2012=1) and are converted to 1997=1 by dividing by the 1997 value of the deflator.

3.2 BLS Capital Variables

NAICS Years: 1997-2018

EQPK, the productive capital stock of equipment, is initially in billions of 2012 dollars. It is converted to billions of 1997 dollars as shown for *EQPK* in Equation (3).

$$EQPK_{i,t} = EQPK_2012B_{i,t} * \frac{PIINVE_2012B_{i,1997}}{100} \quad (3)$$

EQPK_2012B is the initial value of *EQPK* in billions of 2012 dollars and *PIINVE_2012B_i,1997* is the 1997 value of *PIINVE*. The units of the 1997 value of *PIINVE* are (1997$/2012$)*100. *STPK*, the productive capital stock of structures, is converted from billions of 2012 dollars to billions of 1997 dollars in an analogous way. *EQKC*, the capital composition ratio of equipment, is initially given with 2012=1. It is converted to 1997=1 by dividing by the 1997 value of *EQKC*. *STKC*, the capital composition ratio of structures, is converted in an analogous way.

3.3 BEA Capital Variables

BEA provides information on the current-cost net stock of equipment and structures (in current dollars):

- *NKCEQ_CURR*: Current-cost net stock of private equipment and software, year-end estimates ($1B)
- *NKCST_CURR*: Current-cost net stock of private structures, yearend estimates ($1B)

We convert these values into real cost values in billions of 1997 dollar, shown in Equations (4a)-(4b).

$$NKCEQ_{it} = \frac{NKCEQ_CURR_{it}}{PIINVE_{it}} \quad (4a)$$

$$NKCST_{it} = \frac{NKCST_CURR_{it}}{PIINVS_{it}} \quad (4b)$$

where *PIINVE* and *PIINVS* are investment deflators for equipment and structures respectively as described above in Section 3.1.

4. Impute Missing Values in External Industry-Level Data

There are missing values in the NBER-CES database; these are listed below.

- 1992-1997: Asbestos (SIC 3292) – all values [dead/dying industry]
- 1997: SIC industries that move out of manufacturing with switch to NAICS – all values
  - 2411 - Logging
  - 2711 - Newspapers: Publishing, or Publishing and Printing
  - 2721 - Periodicals: Publishing, or Publishing and Printing
  - 2731 – Books: Publishing, or Publishing and Printing
  - 2741 – Miscellaneous Publishing
  - 2771 – Greeting Cards
  - 3732 – Boat Building and Repairing (only part of this industry moved out of manufacturing)
- 2017-2018: The variables *EQUIP* and *PLANT*– all industries
- 2019-2020: all variables for all industries

4.1 Asbestos

Asbestos (SIC 3292) no longer appears in the ASM/CMF publications after 1993. However, the October 2012 version of the NBER-CES database does not contain values for 1992 and 1993. Fortunately, the June 2000 version of the NBER-CES database does contain these values. We use the values for Asbestos in 1992 and 1993 from the June 2000 SIC87 version of the NBER-CES database. Note, we compared Asbestos data from 1958-1991 between the October 2012 SIC and June 2000 SIC87 versions of the NBER-CES databases. The data were an exact match. To convert NBER-CES deflator values to 1997 = 1.0, we estimate Asbestos deflator values out to 1997 in two steps. First, we estimate the annual changes by calculating the activity-weighted average of logarithmic differences in deflators for other 4-digit industries in 329. Then we chain these estimates to the last known levels of the deflator.

4.2 SIC Industries that Drop Out of Manufacturing with the Switch to NAICS in 1997

Several industries move out of manufacturing with the switch to NAICS in 1997: 2411, 2711, 2721, 2731, 2741, 2771, and 3732. Those industries have missing values for all variables for 1997 and later in

the NBER-CES SIC database. We estimate 1997 values of the deflators for the purpose of creating deflator values with 1997=1. We do this using the log change method described in section 4.1. The relevant values are the average of the deflators for other 4-digit industries in the 3-digit industry (for 2731 and 3732) or in the 2-digit industries (for 2411, 2711, 2721, 2741, and 2771).

4.3 2017-2018 NBER-CES Variables

To get 2017-2018 values for the NBER-CES variables of interest, we use the perpetual inventory method. The general method is given by Equation (5) where *K* refers to capital stock and *I* to investment.

$$K_{it+1} = (1 - \delta_{it+1})K_{it} + I_{it+1} \tag{5}$$

Specifically, we impute *EQUIP* (real capital: equipment ($1M)) and *PLANT* (real capital: structures ($1M)) as shown below in Equations (6a) and (6b).

$$EQUIP_{it+1} = (1 - EQDPR_{it+1})EQUIP_{it} + \frac{CME_{it+1}/1000}{PIINVE_{it+1}} \tag{6a}$$

$$PLANT_{it+1} = (1 - STDPR_{it+1})PLANT_{it} + \frac{CBE_{it+1}/1000}{PIINVS_{it+1}} \tag{6b}$$

CME and *CBE* come from the ASM and are divided by 1000 to put them into $1M so they are comparable to *EQUIP* and *PLANT*. *EQDPR* and *STDPR* come from the BLS external data.

4.4 2019-2020 NBER-CES Variables

The NBER-CES data only go through 2018. To get 2019 onward values for NBER-CES variables of interest, we impute data for the missing values. For *EQUIP* and *PLANT* we follow the approach detailed above in section 4.3. For all other variables, we use the method shown in Equation (7) below, where *X* is the variable to be imputed and *Y* is the related variable.

$$X_{it}^{impute} = e^{\log X_{it-1} + \Delta \log Y_{it}} \tag{7}$$

$$\Delta \log Y_{it} = \log(Y_{it}) - \log(Y_{it-1})$$

Table 1 lists the missing variables (*X*) and their corresponding related variables (*Y*).

Table 1. Variables Used to Impute Missing NBER-CES Values

NBER-CES Variable, *X*	Definition	Related Variable, *Y* (Source)
ENERGY	Cost of electricity and fuels ($1M)	*EE* + *CF* (ASM)
EQUIP	Real capital: equipment ($1M)	*CME* (ASM), *EQDPR* (BLS External Data)
MATCOST	Total cost of materials ($1M)	*CP* + *CR* + *CW* + *EE* + *CF* (ASM)
PAY	Total payroll ($1M)	*SW* (ASM)
PIEN	Deflator for ENERGY (1987=1 for SIC; 1997=1 for NAICS)	*PIEN_BEA* (BEA External Data)
PIMAT	Deflator for MATCOST (1987=1 for SIC; 1997=1 for NAICS)	*PIMAT_BEA* (BEA External Data)
PISHIP	Deflator for total value of shipments (1987=1 for SIC; 1997=1.0 for NAICS)	*PISHIP_BEA* (BEA External Data)
PLANT	Real capital: structures ($1M)	*CBE* (ASM), *STDPR* (BLS External Data)

5. Calculate Industry-Level Cost Shares

Total cost for industry *i* in year *t* is calculated as shown in Equation (8).

$$TC_{it} = (EQRKL_{it} * EQUIP_{it}) + (STRKL_{it} * PLANT_{it}) + PAY_{it} + MATCOST_{it} \quad (8)$$

ENERGY is not included in the calculation because it is already included in *MATCOST*. The quantities $EQRKL_{it} = \frac{EQKY_{it}}{EQPK_{it} * EQKC_{it}}$ and $STRKL_{it} = \frac{STKY_{it}}{STPK_{it} * STKC_{it}}$ represent equipment and structure nominal costs of capital. Effectively, we divide capital expenditures *r*K* by *K* to recover *r*. Industry-level cost-shares for year *t* are calculated as shown below in Equations (9) to (13).

Equipment: $$AKE_{it} = \frac{(EQRKL_{it} * EQUIP_{it})}{TC_{it}} \quad (9)$$

Structures: $$AKS_{it} = \frac{(STRKL_{it} * PLANT_{it})}{TC_{it}} \quad (10)$$

Labor: $$AL_{it} = \frac{PAY_{it}}{TC_{it}} \quad (11)$$

Materials: $$AM_{it} = \frac{(MATCOST_{it} - ENERGY_{it})}{TC_{it}} \quad (12)$$

Energy: $$AE_{it} = \frac{ENERGY_{it}}{TC_{it}} \quad (13)$$

Our calculations result in negative cost shares for equipment (*AKE*) and structures (*AKS*) in select years and industries due to negative *EQKY* and *STKY* values in those industry-years.[7] These values come from the BLS external data (roughly at the 2-digit SIC or 3-digit NAICS level). We correct implausible negative values for *EQKY* and *STKY* by interpolating using the average of the *t*-1 and *t*+1 values of *EQKY* and *STKY* for that industry.

5.1 Divisia-Based Cost Shares

Since industry cost shares are noisy, we use Divisia-based cost shares: for year *t* we use the average cost share for *t* and *t*-1. The first year of an industry coding change (1987 and 1997) and the first year of data (1972) are exceptions. For these years, we just use the index in year *t*. Divisia-based industry-level cost shares for year *t* are calculated for all years except 1972, 1987, and 1997 as shown below in Equation (14), where *cs* refers to the relevant cost share among *AKE, AKS, AL, AM,* and *AE.*

$$cs^*_{it} = \frac{1}{2}(cs_{it-1} + cs_{it}) \tag{14}$$

6. Merging External Data to ASM/CMF Data

6.1 ASM/CMF data preparation

The following observations are dropped: administrative record cases in CMF years, observations that are not included in official published statistics, observations with industry codes outside manufacturing, and observations with missing values for payroll, employment, or total value of shipments. Establishment-level longitudinal links are repaired if broken/missing (algorithm not disclosed).

6.2 Merging on SIC External Data (1972-1986)

The 1972-1986 industry-level data are merged to the 1972-1986 ASM/CMF observations by 1972 SIC. As noted in Davis, Haltiwanger, and Schuh (1996), there are miscoded industries in 1972. We correct these problem industry codes using valid 1972 SIC codes for the establishment from 1973-1977, following Davis, Haltiwanger, and Schuh (1996, p. 222). In our correction algorithm we give preference to the year closest to 1972.

6.3 Merging on SIC External Data (1987-1996)

The 1987-1996 industry-level data are merged to the 1987-1996 ASM/CMF observations by 1987 SIC.

6.4 Merging on NAICS External Data (1997-2020)

The 1997-2020 industry-level data are merged to 1997-2020 ASM/CMF observations by 6-digit 2012 NAICS. While minor NAICS changes occur in each of the previous Census years since NAICS codes

[7] *EQKY* and *STKY* are capital income for equipment and structures, respectively. The definition of capital income from BLS is: "Corporate capital income plus imputed noncorporate. Corporate capital income includes corporate capital consumption allowances plus corporate profits before tax without inventory valuation adjustment plus corporate inventory valuation adjustment plus corporate net interest and miscellaneous payments plus corporate business transfer payments plus the part of taxes on production and imports associated with capital (property taxes and motor vehicle taxes). Noncorporate capital income equals total cost less corporate capital income less total labor compensation."

were first implemented in 1997, the 2007/2012 transition saw a major reduction in the number of industries from 473 to 364 6-digit industries. We use a longitudinally consistent version of the 2017 6-digit NAICS for our calculations of total factor productivity and the creation of the DiSP product, but we include additional vintages of SIC and NAICS codes in the microdata for researchers who prefer to take a different approach.

All ASM/CMF observations with valid manufacturing NAICS codes successfully merge to the industry-level data by year and 6-digit NAICS.

7. Calculate Establishment-Level Inputs and Output

This section describes the calculation of establishment-level inputs (capital, labor, energy, and materials) and output. Table 2 shows the variables used in the calculations. ASM/CMF variables are establishment-level. Other variables are industry-level variables as described above and in Appendix A.

Table 2. Variables Used to Calculate Establishment-Level Inputs and Output

Variable	Definition
BAE	Building and other structures assets at the end of the year ($K) [ASM/CMF]
CBE	Capital expenditures on new and used buildings and other structures ($K) [= *NB* + *UB*] [ASM/CMF]
CF	Cost of fuels ($K) [ASM/CMF]
CME	Capital expenditures on new and used machinery and equipment ($K) [= *NM* + *UM*] [ASM/CMF]
CP	Cost of materials and parts ($K) [ASM/CMF]
CR	Cost of resales ($K) [ASM/CMF]
CW	Cost of contract work ($K) [ASM/CMF]
EE	Cost of purchased electricity ($K) [ASM/CMF]
EQDPR	Depreciation rate of wealth stock, equipment [BLS]
FIB	Finished goods inventory beginning ($K) [ASM/CMF]
FIE	Finished goods inventory ending ($K) [ASM/CMF]
GKHEQ	Historical-cost net stock of private equipment and software, yearend estimates ($B) [Industry-level, BEA]
GKHST	Historical-cost net stock of private structures, yearend estimates ($B) [Industry-level, BEA]
MAE	Machinery and equipment assets at the end of the year ($K) [ASM/CMF]
NB	Capital expenditures on new buildings and other structures ($K) [ASM/CMF]
NKCEQ	Current-cost net stock of private equipment and software, yearend estimates (1997 $B) [Industry-level, Calculated from BEA and BLS]
NKCST	Current-cost net stock of private structures, yearend estimates (1997 $B) [Industry-level, Calculated from BEA and BLS]
NM	Capital expenditures on new machinery and equipment ($K) [ASM/CMF]
PH	Production worker hours (1000 hours) [ASM/CMF]
PIEN	Deflator for energy (1997=1.0) [NBER-CES]
PIINVE	Investment price deflator, equipment (1997=1.0) [BLS]
PIINVS	Investment price deflator, structures (1997=1.0) [BLS]
PIMAT	Deflator for materials (1997=1.0) [NBER-CES]

PISHIP	Deflator for shipments (1997=1.0) [NBER-CES]
PW	Average production worker hours [ASM/CMF]
STDPR	Depreciation rate of wealth stock, structures [BLS]
SW	Total salaries and wages ($K) [ASM/CMF]
TVS	Total value of shipments ($K) [ASM/CMF]
UB	Capital expenditures on used buildings and other structures ($K) [ASM/CMF]
UM	Capital expenditures on used machinery and equipment ($K) [ASM/CMF]
WIB	Work-in-process inventory beginning ($K) [ASM/CMF]
WIE	Work-in-process inventory ending ($K) [ASM/CMF]
WW	Production worker wages ($K) [ASM/CMF]

7.1 Capital

Capital Stock and Expenditures Questions on the ASM Through Time

The ASM requests information on the gross value of depreciable assets broken out by "buildings and other structures" and "machinery and equipment" from 1972-1985. Starting in 1986, the ASM does not ask about the gross value of depreciable assets unless it is a CMF year. The CMF requests asset information broken out by "buildings and other structures" and "machinery and equipment" in 1987 and 1992. Starting in 1997, only information on the total gross value of depreciable assets is requested.

From 1972-1976, the ASM requests information on three categories of capital expenditures: new buildings and other structures; new machinery and equipment; and used buildings and other structures together with machinery and equipment. Starting in 1977, the ASM requests information on used buildings and other structures and used machinery and equipment separately. In 1997, the ASM switches to request only total new and used expenditures on buildings and other structures and total new and used expenditures on machinery and equipment. This continues through the present.

Capital Stock Calculations: Initialization

We use the perpetual inventory method to calculate capital stocks where possible. Given there are breaks in data for individual establishments and establishments that only appear once in the data (e.g., many CMF non-ASM establishments), this is not always possible.

Method 1

From 1972-1985 and for CMF years 1987 and 1992, we have reported book value of capital stock split out by machinery and equipment and buildings and other structures:

- *MAE* = machinery and equipment assets at the end of the year (1000 current dollars)
- *BAE* = building and other structures assets at the end of the year (1000 current dollars)

In these years, we can calculate initial real capital stock for machinery and equipment and for buildings and structures as shown in Equations (15a) and (15b).

$$KSTOCK_EQ_{et} = MAE_{et} * \left(\frac{NKCEQ_{it}}{GKHEQ_{it}}\right) \quad (15a)$$

$$KSTOCK_ST_{et} = BAE_{et} * \left(\frac{NKCST_{it}}{GKHST_{it}}\right) \quad (15b)$$

However, upon investigating the properties of the aggregate time series, we noted that, in the early period of the CMP dataset in particular, our capital stocks overstate the size of the capital stock relative to the BEA. Using the sum of reported values of each capital class for all available establishments within a BEA sector in any year where initialization using reported stocks was possible, and the BEA-reported historical-value capital stock, we can create a ratio between reported capital in CMP and BEA sectoral capital. We denote this adjustment factor as *EQ_ADJ* for equipment and *ST_ADJ* for structures for each industry *i*. The formula for materials and equipment is given below in Equations (16a) and (16b).

$$EQ_ADJ_{it} = \left(\frac{\sum_{e \in i} MAE_{et}}{GKHEQ_{it} * PIINVE_{it}}\right) \quad (16a)$$

$$ST_ADJ_{it} = \left(\frac{\sum_{e \in i} BAE_{et}}{GKHST_{it} * PIINVS_{it}}\right) \quad (16b)$$

We then adjust the individual reported values by this factor, depressing the values of reported capital. This yields the following capital stock in an industry for establishment *e* in industry *i* at time *t*:

$$KSTOCK_EQ_{et} = MAE_{et} * EQ_ADJ_{it} * \left(\frac{NKCEQ_{it}}{GKHEQ_{it}}\right) \quad (17a)$$

$$KSTOCK_ST_{et} = BAE_{et} * ST_ADJ_{it} * \left(\frac{NKCST_{it}}{GKHST_{it}}\right) \quad (17b)$$

Note, *NKCEQ* and *NKCST* are in billions of 1997 dollars, *GKHEQ* and *GKHST* are in billions of current dollars, and *MAE* and *BAE* are in thousands of current dollars. Thus, calculated *KSTOCK_EQ* and *KSTOCK_ST* are in thousands of 1997 dollars.

Method 2

Starting with the 1997 CM, we only have information on beginning and end of year total assets (*TAE* and *TAB*). However, we do have investment information split out by machinery and equipment and buildings and other structures. For establishments that appear for the first time in 1993 or later and appear in the 1997, 2002, 2007, 2012, or 2017 CM, we use a combination of the following two imputation methods to initialize capital. First, we need the same adjustment factor from method 1, but for total assets:

$$TAE_ADJ_{it} = \left(\frac{\sum_{e \in i} TAE_{et}}{(GKHEQ_{it} * PIINVE_t + GKHST_{it} * PIINVS_{it})}\right) \quad (18)$$

Imputation 1: Use industry-level shares.

$$KSTOCK_EQ_1_{et} = KSHARE_EQ_I_{it} * TA_ADJ_{it} * TAE_{et} * \left(\frac{NKCEQ_{it}}{GKHEQ_{it}}\right) \quad (19a)$$

$$KSTOCK_ST_1_{et} = KSHARE_ST_I_{it} * TA_ADJ_{it} * TAE_{et} * \left(\frac{NKCST_{it}}{GKHST_{it}}\right) \quad (19b)$$

where *KSHARE_EQ_1* and *KSHARE_ST_1* are industry-year level shares calculated as shown:

$$KSHARE_EQ_1_{it} = \frac{EQUIP_{it}}{(EQUIP_{it}+PLANT_{it})} \tag{20a}$$

$$KSHARE_ST_1_{it} = \frac{PLANT_{it}}{(EQUIP_{it}+PLANT_{it})} \tag{20b}$$

Imputation 2: Use establishment-level investment flow shares.

$$KSTOCK_EQ_2_{et} = KSHARE_EQ_P_{et} * TA_ADJ_{it} * TAE_{et} * \left(\frac{NKCEQ_{it}}{GKHEQ_{it}}\right) \tag{21a}$$

$$KSTOCK_ST_2_{et} = KSHARE_ST_P_{et} * TA_ADJ_{it} * TAE_{et} * \left(\frac{NKCST_{it}}{GKHST_{it}}\right) \tag{21b}$$

where *KSHARE_EQ_2* and *KSHARE_ST_2* are establishment-year level shares calculated as shown:

$$KSHARE_EQ_P_{et} = \frac{\sum_t \frac{CME_{et}}{PIINVE_{it}}}{\sum_t \frac{CME_{et}}{PIINVE_{it}} + \sum_t \frac{CBE_{et}}{PIINVS_{it}}} \tag{22a}$$

$$KSHARE_ST_P_{et} = \frac{\sum_t \frac{CBE_{et}}{PIINVS_{it}}}{\sum_t \frac{CME_{et}}{PIINVE_{it}} + \sum_t \frac{CBE_{et}}{PIINVS_{it}}} \tag{22b}$$

where we are summing across all years 1997 and later in which establishment *e* has data.

Note that $KSHARE_EQ_2_{et}$ and $KSHARE_ST_2_{et}$ contain deflators because we are summing across years, whereas to calculate $KSHARE_EQ_1_{et}$ and $KSHARE_ST_1_{et}$ deflators are not needed because we are only using data from a single year.

We use a regression model to determine the best linear weight for these two methods over the 1972-1985, 1987, 1992 period. The final formula for cases where both industry and establishment-level ratios exist is given by

$$KSTOCK_EQ_{et} = indwt_i KSTOCK_EQ_1_{et} + (1 - indwt_i) KSTOCK_EQ_2_{et} \tag{23a}$$

$$KSTOCK_ST_{et} = indwt_i KSTOCK_ST_1_{et} + (1 - indwt_i) KSTSTOCK_2_{et} \tag{23b}$$

where *indwt* is the weight put on the industry value and is between 0 and 1.

We prioritize imputation method 2, at the establishment level, but use imputation method 1, at the industry level, in cases where establishment-level information is not available.[8]

Method 3

For establishment-year observations for which we cannot calculate initial capital stock using any of the above procedures and for which we cannot calculate capital stock using either forward or backward

[8] Details about the calculation of *indwt* are available in CMP Team (2023).

perpetual inventory method, we follow an approach similar to that used by Bloom et al. (2013). The method suggests we impute the initial capital stock for the establishment using the ratio of investment to the book value of capital stock (I/K ratio) in the prior CMF. However, we have found that using revenue to capital ratios (Q/K) are more stable and produce a more "well-behaved" aggregate series.[9] Establishment-level Q/K ratios can be calculated as in Equations (24a) and (24b):[10]

$$RATIO_Q_K_EQ_{et} = \frac{Q_{et}}{KSTOCK_EQ_{et}} \tag{24a}$$

$$RATIO_Q_K_ST_{et} = \frac{Q_{et}}{KSTOCK_ST_{et}} \tag{24b}$$

- If the establishment is in the prior CMF, we use the establishment's Q/K ratio from that CMF.
- If the establishment is not in the prior CMF, we use an industry ratio. We calculate industry-level ratios as the mean of the establishment-level ratios.[11]
    - For young establishments (those 0-4 years of age), we apply a separate industry-level Q/K ratio calculated using only young establishments.
    - For establishments that are not young, we apply an industry-level Q/K ratio calculated using all establishments in the industry.

The calculation of capital stock using the Q/K ratio is shown below, where the *e (or i)* subscript indicates that either an establishment- or industry-level ratio may be used.

$$KSTOCK_EQ_{et} = \frac{Q_{et}}{RATIO_Q_K_EQ_{e(or\ i),t}} \tag{25a}$$

$$KSTOCK_ST_{et} = \frac{Q_{et}}{RATIO_Q_K_ST_{e(or\ i),t}} \tag{25b}$$

Capital Stock Calculations: Perpetual Inventory Method

Once capital stock is initialized, we calculate real capital stock for future years using perpetual inventory as shown in Equations (26a) and (26b) where possible.

$$KSTOCK_EQ_{et} = (1 - EQDPR_{it}) * KSTOCK_EQ_{et-1} + \left(\frac{CME_{et}}{PIINVE_{it}}\right) \tag{26a}$$

$$KSTOCK_ST_{et} = (1 - STDPR_{it}) * KSTOCK_ST_{et-1} + \left(\frac{CBE_{et}}{PIINVS_{it}}\right) \tag{26b}$$

where

- *CME* = capital expenditures on machinery and equipment (1000 current dollars)

[9] We always go back to the prior CMF thereby avoiding the possibility of running into an industry code system change. In future work, we will explore the idea of using the nearest CMF.

[10] More details about the capital stock initialization methodology, and comparisons with the original I/K method are available in CMP Team (2023).

[11] We take the mean of the establishment-level ratios. One could alternatively take the mean of Q and the mean of K and then divide, as all the underlying data are on the dataset.

- *CBE* = capital expenditures on buildings and other structures (1000 current dollars)

Prior to 2002, *CME* and *CBE* are not on the ASM and CMF files, but they can be easily calculated for most years. In the years 1982, 1987, and 1989-1996, *CME* and *CBE* can be calculated as shown below in Equations (27a) and (27b). For 1997-2001, *NB* and *NM* are incorrectly populated with what is actually the sum of new and used values (only the sum was collected in 1997-2001). Future versions of the ASM may correct this issue. In the meantime, we rename *NB* and *NM* to *CBE* and *CME* in 1997-2001.

$$CME_{et} = NM_{et} + UM_{et} \tag{27a}$$

$$CBE_{et} = NB_{et} + UB_{et} \tag{27b}$$

Prior to 1982, and in the years 1983-1986 and 1988, we do not have used capital expenditures broken out by machinery and equipment. In those years, we estimate *CME* and *CBE* by utilizing the split between machinery and equipment for new capital expenditures in industry-level regressions to capture the relationship between used and new expenditure splits. Specifically, we estimate the following regression in the years 1987, 1989-1996:[12]

$$UB_Share_{et} = \alpha_i + \beta_i \, NB_Share_{et} + \epsilon_{et} \tag{28a}$$

where

$$UB_Share_{et} = UB_{et}/(UB_{et} + UM_{et}) \tag{28b}$$

$$NB_Share_{et} = NB_{et}/(NB_{et} + NM_{et}) \tag{28c}$$

Since this regression is run at the industry level, α_i gives an industry fixed effect for the splits between used structures and used equipment. The relationship for industry *i* between the used and new share in these years is given by β_i. Using estimates $\hat{\alpha}_i$ and $\hat{\beta}_i$, we impute the split between used buildings and used expenditures in 1972-1981, 1983-1986, and 1988 according to the share of buildings predicted by the above regressions.

$$\widehat{UB_Share}_{et} = \hat{\alpha}_i + \hat{\beta}_i \, NB_Share_{et} \tag{29a}$$

$$\widehat{UB}_{et} = UE_{et} * \widehat{UB_Share}_{et} \tag{29b}$$

$$\widehat{UM}_{et} = UE_{et} * (1 - \widehat{UB_{Share}}_{et}) \tag{29c}$$

There is a potential conflict created by using the perpetual inventory method. For example, say we have capital stock in 1987 and 1992. If we initialize capital stock in 1987 using Equations (17a) and (17b) and then use (26a) and (26b) to calculate capital stock in 1988-1992, we will probably get a different value in 1992 than what is reported. In this case, we have decided to ignore the 1992 reported data and keep the series consistent. In future work, alternative options may be explored.

[12] Note that we do not use 1982, since it had similar issues to the ASM years surrounding it.

For establishments that appear for the first time in the ASM, we do not have an initial value for their capital stock. For these establishments, we go backwards from the most recent CMF as shown in (30a) and (30b).

$$KSTOCK_EQ_{et} = \left(KSTOCK_EQ_{e,t+1} - \left(\frac{CME_{e,t+1}}{PIINVE_{i,t+1}}\right)\right)/\left(1 - EQDPR_{i,t+1}\right) \quad (30a)$$

$$KSTOCK_ST_{et} = \left(KSTOCK_ST_{e,t+1} - \left(\frac{CBE_{e,t+1}}{PIINVS_{i,t+1}}\right)\right)/\left(1 - STDPR_{i,t+1}\right) \quad (30b)$$

Algorithm for Assigning Capital Values to Observations

Below is a summary of the sequence of assigning a value to the capital stock measure. FWD-PIM and BWD-PIM denote a value from a forward and backward iteration of the perpetual inventory method. We use the term 'gap year' to label instances where either the current or the previous (subsequent) or both years are absent from the original panel. Q/K-PIM denotes our method 3 for capital initialization.

Regular PIM

1. If the former year is not a gap year, apply FWD-PIM. If no value is assigned, try to initialize using book value.
2. If the subsequent year is not a gap year and no value was assigned in FWD-PIM, apply BWD-PIM. If no value is assigned, try to initialize using book value.

Q/K-initialization

3. Initialize and apply FWD Q/K-PIM on original dataset with possibly discontinuous time series. Restrictions are the same as for FWD-PIM.
4. If a value was not assigned in FWD Q/K-PIM, apply BWD Q/K-PIM. Restrictions are the same as for BWD-PIM

7.2 Labor

Labor is given by total hours calculated as shown in Equation (31) when $SW > 0$ and $WW > 0$.[13] When SW or WW are zero, we instead calculate total hours as simply $TH_{et} = PH_{et}$.

$$\text{If } SW_{et} > 0 \text{ and } WW_{et} > 0, \text{ then } TH_{et} = (PH_{et} * SW_{et})/WW_{et} \quad (31)$$

We also include an alternative measure of labor using data from the BLS. Using the CPS, they calculate the ratio of nonproduction to production worker hours (*CPSRatio*) at the 4-digit NAICS level. We use this ratio to adjust total hours, as in Equation (32), when $SW_{et} > 0$ and $WW_{et} > 0$.

$$TH_BLS_{et} = PH_{et}\left(1 + \left(\frac{1}{PW_{et}}\right) * (TE_{et} - PW_{et}) * CPSRatio_{it}\right) \quad (32)$$

[13] Note, Kehrig (2011) proposes an alternate measure for *TH*, as well as alternatives for energy costs, materials, and output. We do not use these methods here, although all the necessary inputs are available on the dataset for researchers to construct those measures.

7.3 Energy

Real establishment-level energy costs are calculated as:

$$E_{et} = (EE_{et} + CF_{et})/PIEN_{it} \quad (33)$$

7.4 Materials

Real establishment-level materials costs are calculated as:

$$M_{et} = (CP_{et} + CR_{et} + CW_{et})/PIMAT_{it} \quad (34)$$

7.5 Total Output

Real establishment-level total output is calculated as:

$$Q_{et} = (TVS_{et} + DF_{et} + DW_{et})/PISHIP_{it} \quad (35)$$

where $DF_{et} = FIE_{et} - FIB_{et}$ and $DW_{et} = WIE_{et} - WIB_{et}$. When components of *DF* or *DW* are missing, they are set to zero. When the resulting Q_{et} is negative, we instead calculate total output as $Q_{et} = TVS_{et}/PISHIP_{it}$.

8. Establishment-Level TFP

We calculate establishment-level log TFP as shown in Equation (36).

$$\begin{aligned} LTFP_{et} = \log(Q_{et}) - IAKE_{it} * \log(KSTEQ_{et}) - IAKS_{it} * \log(KSTST_{et}) \\ -IAL_{it} * \log(TH_{et}) - IAM_{it} * \log(M_{et}) - IAE_{it} * \log(E_{et}) \end{aligned} \quad (36)$$

We only calculate *LTFP* for establishments with positive values for each of the establishment-level inputs and output.

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Appendix A. External Data

This appendix describes the various public, industry-level external datasets used in the construction of the establishment-level TFP dataset. External data are obtained from three main sources:

- NBER-CES Manufacturing Industry Database
- Bureau of Economic Analysis (BEA)
- Bureau of Labor Statistics (BLS)

Tables A-1a through A-1c provide summary information for the external data sources. Tables A-2 through A-4 list the relevant variables read in from the external data sources.

There are some missing values in the NBER-CES Manufacturing Industry Database. Those that are relevant for construction of an establishment-level TFP dataset are discussed in the notes after Table A-4.

Table A-1a. External Data Sources and Summary Information, NBER-CES

NBER-CES Manufacturing Industry Database (March 2021 Version)	
Website	www.nber.org/nberces/
Years	1958-2018
Notes	See Table A-2 for more information on the NBER-CES data.

Table A-1b. External Data Sources and Summary Information, BEA

Bureau of Economic Analysis (BEA) – NAICS	
Website	www.bea.gov
Years	1947-2020
Notes	BEA provides only industry descriptions. These descriptions were converted into NAICS codes (roughly NAICS3, but some NAICS3 and NAICS4 groups) using a concordance provided in "Local Area Personal Income and Employment Methodology" (September 2019) [https://www.bea.gov/system/files/methodologies/LAPI-Methodology.pdf] We use the BEA series: "Current-Cost Net Stock of Private Equipment by Industry", "Current-Cost Net Stock of Private Structures by Industry", "Historical-Cost Net Stock of Private Equipment by Industry", and "Historical-Cost Net Stock of Private Structures by Industry" from tables 3.1E, 3.1S, 3.3E, and 3.3S. [https://apps.bea.gov/iTable/?isuri=1&reqid=10&step=4&categories=flatfiles&nipa_table_list=1]. We also use the BEA series: "Chain-Type Price Indexes for Energy Inputs by Industry", "Chain-Type Price Indexes for Materials Inputs by Industry", and "Chain-Type Price Indexes for Gross Output by Industry" [https://apps.bea.gov/iTable/iTable.cfm?isuri=1&reqid=151&step=1; data for energy and materials come from KLEMS; data for gross output come from GrossOutput]. Historical chain-type price indices for energy, materials, and gross output come from the "Annual Table" [https://apps.bea.gov/iTable/iTable.cfm?reqid=147&step=2&isuri=1].
Bureau of Economic Analysis (BEA) - SIC	
Years	1971-2000
Notes	Raw data files obtained by Hyowook Chiang BEA provides only industry descriptions. These descriptions were converted into SIC codes (roughly SIC2, but some SIC3 groups) using a concordance provided in "Local Area Personal Income and Employment Methodology" (September 2019) [https://www.bea.gov/system/files/methodologies/LAPI-Methodology.pdf]

Table A-1c. External Data Sources and Summary Information, BLS

Bureau of Labor Statistics (BLS) - NAICS	
Website	www.bls.gov
Years	1987-2020
Notes	BLS provides these data at the 3-digit NAICS level with some 3-digit NAICS combined. We use the BLS Capital Details for Major Sectors and Industries, 1987-2020 [https://www.bls.gov/productivity/tables/home.htm under "Total Factor Productivity and Related Measures"]
Bureau of Labor Statistics (BLS) - SIC	
Years	1971-2000
Notes	These data are at the 2-digit SIC level. Raw data files obtained by Hyowook Chiang.

Table A-2. External Variables: NBER-CES Database

Variable Name	Available Years	Industry Type	Definition
NBER-CES Manufacturing Industry Database (March 2021 Version)			
ENERGY	1958-2018	87 SIC4, 12 NAICS6	Cost of electricity and fuels ($1M)
EQUIP	1958-2016	87 SIC4, 12 NAICS6	Real capital: equipment ($1M)
MATCOST	1958-2018	87 SIC4, 12 NAICS6	Total cost of materials ($1M)
PAY	1958-2018	87 SIC4, 12 NAICS6	Total payroll ($1M)
PIEN	1958-2018	87 SIC4, 12 NAICS6	Deflator for ENERGY (1987 = 1.0 for SIC; 1997 = 1.0 for NAICS)
PIMAT	1958-2018	87 SIC4, 12 NAICS6	Deflator for MATCOST (1987 = 1.0 for SIC; 1997 = 1.0 for NAICS)
PISHIP	1958-2018	87 SIC4, 12 NAICS6	Deflator for total value of shipments (1987 = 1.0 for SIC; 1997 = 1.0 for NAICS)
PLANT	1958-2016	87 SIC4, 12 NAICS6	Real capital: structures ($1M)
NBER-CES Manufacturing Industry Database			
ENERGY	1958-1996	72 SIC4, 87 SIC4	Cost of electricity and fuels ($1M)
EQUIP	1958-1996	72 SIC4, 87 SIC4	Real capital: equipment ($1M)
MATCOST	1958-1996	72 SIC4, 87 SIC4	Total cost of materials ($1M)
PAY	1958-1996	72 SIC4, 87 SIC4	Total payroll ($1M)
PIEN	1958-1996	72 SIC4, 87 SIC4	Deflator for ENERGY (1987 = 1.0)
PIMAT	1958-1996	72 SIC4, 87 SIC4	Deflator for MATCOST (1987 = 1.0)
PISHIP	1958-1996	72 SIC4, 87 SIC4	Deflator for total value of shipments (1987 = 1.0)
PLANT	1958-1996	72 SIC4, 87 SIC4	Real capital: structures ($1M)

Table A-3. External Variables: Bureau of Economic Analysis

Variable Name	Available Years	Industry Type	Definition
Bureau of Economic Analysis - NAICS			
GKHEQ	1947-2020	BEA-defined[1]	Historical-cost net stock of private equipment and software, yearend estimates ($1B) [Table 3.3E]
GKHST	1947-2020	BEA-defined[1]	Historical-cost net stock of private structures, yearend estimates ($1B) [Table 3.3S]
PIEN_BEA_2012B	1997-2020	BEA-defined[1]	Chain-type price indexes for energy inputs (2012=100) [GDP-by-Industry, KLEMS tables]
PIMAT_BEA_2012B	1997-2020	BEA-defined[1]	Chain-type price indexes for materials inputs (2012=100) [GDP-by-Industry, KLEMS tables]
PISHIP_BEA_2012B	1987-2020	BEA-defined[1]	Chain-type price indexes for gross output (2012=100) [GDP-by-Industry, Gross Output by Industry tables]
NKCEQ_CURR	1947-2020	BEA-defined[1]	Current-cost net stock of private equipment and software, yearend estimates ($1B) [Table 3.1E]
NKCST_CURR	1947-2020	BEA-defined[1]	Current-cost net stock of private structures, yearend estimates ($1B) [Table 3.1S]
Bureau of Economic Analysis – SIC[2]			
GKHEQ	1972-1997	BEA-defined[3]	Historical-Cost Net Stock: Equipment ($1B) [Table 3.3E]
GKHST	1972-1997	BEA-defined[3]	Historical-Cost Net Stock: Structures ($1B) [Table 3.3S]
NKCEQ_CURR	1972-1997	BEA-defined[3]	Current-cost net stock of private equipment and software, yearend estimates ($1B) [Table 3.1E]
NKCST_CURR	1972-1997	BEA-defined[3]	Current-cost net stock of private structures, yearend estimates ($1B) [Table 3.1S]

[1] BEA provides only NAICS-based industry descriptions. These descriptions were converted into NAICS codes (roughly NAICS3, but some NAICS3 and NAICS4 groups) using a concordance provided in "Local Area Personal Income and Employment Methodology" [https://www.bea.gov/system/files/methodologies/LAPI-Methodology.pdf].

[2] For consistency with the NAICS-based data, we convert these values from millions of current dollars to billions of current dollars when we read them in.

[3] BEA provides only SIC-based industry descriptions. These descriptions were converted into SIC codes (roughly SIC2, but some SIC3 groups) using a concordance provided in "Local Area Personal Income and Employment Methodology" [https://www.bea.gov/system/files/methodologies/LAPI-Methodology.pdf]

Table A-4. External Variables: Bureau of Labor Statistics

Variable Name	Available Years	Industry Type	Definition
Bureau of Labor Statistics - NAICS			
EQDPR	1987-2020	NAICS3[1]	Depreciation rate of wealth stock, equipment [Table 9-1]
EQKC_2012B	1987-2020	NAICS3[1]	Capital composition (ratio of capital input to productive stock), equipment (2012 = 100) [Table 6-2]
EQKY	1987-2020	NAICS3[1]	Capital income in billions of current dollars, equipment [Table 3-1]
EQPK_2012B	1987-2020	NAICS3[1]	Productive capital stock (direct aggregates in billions of 2012 dollars), equipment [Table 4-1]
PIINVE_2012B	1987-2020	NAICS3[1]	Investment price deflator (2012 = 100), equipment [Table 8-2]
PIINVS_2012B	1987-2020	NAICS3[1]	Investment price deflator (2012 = 100), structures [Table 8-2]
STDPR	1987-2020	NAICS3[1]	Depreciation rate of wealth stock, structures [Table 9-1]
STKC_2012B	1987-2020	NAICS3[1]	Capital composition (ratio of capital input to productive stock), structures [Table 6-2]
STKY	1987-2020	NAICS3[1]	Capital income in billions of current dollars, structures [Table 3-1]
STPK_2012B	1987-2020	NAICS3[1]	Productive capital stock (direct aggregates in billions of 2012 dollars), structures [Table 4-1]
Bureau of Labor Statistics - SIC			
EQDPR	1948-2000	SIC2	Depreciation rate of wealth stock, equipment [Table 9a]
EQKC_1996B	1948-2000	SIC2	Capital composition (ratio of capital input to productive stock), equipment (1996 = 100) [Table 6b]
EQPK_1996B	1948-2000	SIC2	Productive capital stock (direct aggregates in billions of 1996 dollars), equipment [Table 4a]
EQKY	1948-2000	SIC2	Capital income in billions of current dollars, equipment [Table 3a]
PIINVE_1996B	1948-2000	SIC2	Investment price deflator (1996 = 100), equipment [Table 8b]
PIINVS_1996B	1948-2000	SIC2	Investment price deflator (1996 = 100), structures [Table 8b]
STDPR	1948-2000	SIC2	Depreciation rate of wealth stock, equipment [Table 9a]
STKC_1996B	1948-2000	SIC2	Capital composition (ratio of capital input to productive stock), equipment (1996 = 100) [Table 6b]
STPK_1996B	1948-2000	SIC2	Productive capital stock (direct aggregates in billions of 1996 dollars), equipment [Table 4a]
STKY	1948-2000	SIC2	Capital income in billions of current dollars, equipment [Table 3a]

[1] BLS provides information at the 3-digit NAICS level with some 3-digit NAICS combined.